Push 2 / Nuendo Bridge

User Guide & Installation Manual — Version 1.0.8

Compatible with Nuendo 14+ and Cubase 14+ — macOS and Windows

This guide covers installation, configuration, and use of the Push 2 / Nuendo Bridge.

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1. System Requirements

- Ableton Push 2 (USB), Steinberg Nuendo 14+ or Cubase 14+, macOS 11+ or Windows 10/11
- **Standalone app/exe: nothing else to install.** The macOS .app and Windows .exe bundle the Python interpreter, all dependencies and the libusb USB runtime. No Homebrew, no libusb, no Python.
- **Running from source only:** Python 3.9–3.11.9 (3.12/3.13 not yet supported), libusb (macOS: `brew install libusb`), and push2-python from source:
  `pip install git+https://github.com/ffont/push2-python.git`
- Plugin Mapper (optional): pip install fastapi uvicorn pedalboard

  Nuendo 15+ features: *Plugin Browser, DirectAccess insert control and the Channel Strip extended-parameter / EQ-curve pages require API 1.3. All other features work with Nuendo 14+.*

2. Installation — macOS

Option A: Standalone App — Apple Silicon (Recommended)

*The pre-built .app is **Apple Silicon (arm64) only**. On an M1/M2/M3 Mac it bundles everything (Python, all dependencies and the libusb USB runtime) — no Homebrew, libusb or Python required. **Intel Macs: see Option B below.***

- Step 1: Copy Push2 Nuendo Bridge.app to /Applications
- Step 2: The app is **not code-signed**, so macOS Gatekeeper blocks it on first launch. Open Terminal and remove the quarantine flag (match the version in the .app name to your copy):

```
xattr -dr com.apple.quarantine "/Applications/Push2 Nuendo Bridge_v1.0.8.app"
```

- Step 3: Copy Ableton_Push2.js to:
  ~/Documents/Steinberg/Nuendo/MIDI Remote/Driver Scripts/Local/Ableton/Push2/
  Create the Ableton/Push2 folder if it doesn't exist.
- Step 4: Double-click the app. A P2 icon appears in the menu bar.

Option B: Run from Source — Intel Macs

The pre-built .app does not run on Intel Macs. Run from source instead (this also builds an Intel-native .app if you prefer — see the note at the end). Open Terminal:

- Step 1: Install Homebrew if needed — see https://brew.sh
- Step 2: Install libusb and Python 3.11 — `brew install libusb python@3.11`
- Step 3: From the project folder, install the dependencies:

```
pip3.11 install -r requirements.txt
pip3.11 install git+https://github.com/ffont/push2-python.git
```

- Step 4 (optional, Plugin Mapper): `pip3.11 install fastapi uvicorn pedalboard`
- Step 5: Copy Ableton_Push2.js to the MIDI Remote folder (see Option A, Step 3).
- Step 6: Run the bridge — `cd src && python3.11 main.py`

  To build a standalone Intel .app on an Intel Mac, run `bash scripts/build.sh` *(with the dependencies above installed). PyInstaller produces an architecture-native app, so building on Intel yields an Intel .app.*

3. Installation — Windows

Windows now ships as a **standalone .exe** — no Python, pip or command line required. See the separate Windows Installation Guide for the step-by-step walkthrough. Key points:

- Download **Push2NuendoBridge-vX.Y.Z-Windows.zip** and unzip it anywhere (e.g. Documents). It contains the `.exe`, `Ableton_Push2.js` and the PDF guides.
- Install **loopMIDI** and create the four ports: `NuendoBridge In`, `NuendoBridge Out`, `BridgeNotes`, `BridgeNotes In`. Set it to start with Windows.
- **Plug in the Push 2** — Windows installs the required USB driver (WinUSB) automatically the first time you connect it. No Zadig, no manual driver step.

  Note: the bridge and Ableton Live share the same Push 2 USB connection, so they cannot both control it at the same time — close one to use the other. macOS is not affected. (If Windows ever fails to auto-install the driver, the Windows Installation Guide's Troubleshooting section shows how to force it with Zadig.)
- **Double-click the .exe** to run the bridge — it starts in the **system tray** (icon near the clock), not a console window. Right-click the icon for status, Open Plugin Mapper (http://localhost:8100), Show Log and Quit. Run with `--terminal` for the classic console version.

  No Python install is needed: the interpreter, all dependencies and the libusb runtime are bundled inside the .exe.
- Copy **Ableton_Push2.js** to the MIDI Remote scripts folder, creating the sub-folders if needed:

```
Nuendo:
C:\Users\<user>\Documents\Steinberg\Nuendo\MIDI Remote\Driver
Scripts\Local\Ableton\Push2\
```

```
Cubase:
C:\Users\<user>\Documents\Steinberg\Cubase\MIDI Remote\Driver
Scripts\Local\Ableton\Push2\
```

  Replace <user> with your Windows user name. The `Local\Ableton\Push2` folders usually do not exist yet — create them.

4. First Launch

Connect Push 2 via USB, launch the bridge, open Nuendo. Menu bar icon: P2 ✓ = connected, P2 ■ = waiting. Click for Start/Stop, Show Logs, Plugin Mapper, Start at Login, Quit.

5. Nuendo Configuration

Automatic setup: Nuendo usually detects the script and assigns its ports on its own.

If the script does not load automatically

- Open any project.
- Studio → MIDI Remote Manager → **Add MIDI Controller Surface**.
- Vendor = **Ableton**, Model = **Push2**.
- MIDI Input = **NuendoBridge Out**, MIDI Output = **NuendoBridge In** (the ports carry the same names as listed here).

Note Input

Set the instrument track's MIDI input to **BridgeNotes** (enable it in MIDI Port Setup if hidden).

Important — avoid doubled notes

- In Studio Setup → MIDI Port Setup, **uncheck “In ‘All MIDI Inputs’” for both Push 2 ports** — “Ableton Push 2” (Live Port) and “Ableton Push 2 User Port”.

  If left checked, every pad you play is received twice: once via the bridge's BridgeNotes port and once directly from the Push 2 hardware. Excluding the raw Push 2 ports is standard practice for any control-surface bridge and is permanent once set.

- Also **uncheck “In ‘All MIDI Inputs’” for NuendoBridge Out**. This port carries the bridge's control messages to the MIDI Remote script; if it is part of All MIDI Inputs, MIDI tracks receive those messages too and their parameters can change unexpectedly. Since 1.0.8 the setting is kept between launches (on macOS, uncheck it once more after upgrading — the port appears as new the first time).

6. Using the Bridge

8 vertical zones on the Push 2 screen. Left/Right = bank by 8 tracks. Shift+Left/Right = nudge by 1 track (Volume/Pan modes).

Button	Mode	Encoders
Mix	Volume	8 track volumes
Mix (again)	Channel Strip	Strip module params
Shift+Mix	Track	Vol+Pan+Sends
Clip	Sends	8 sends
Clip (again)	Pan	8 track pans
Device	Quick Controls	8 QC
Browse	Inserts	Plugin params
Add Device	Plugin Browser	Browse/load plugins
Session	XY pad	Morph 2 track params / CC
Master	Navigation	D-pads: zoom / scroll / markers / nudge

Lower row: Mute toggles Mute/Monitor. Solo toggles Solo/Rec. Shift+Mute/Solo clears all.

Long-press menus

Hold (≈half a second) **New, Duplicate, Mute, Play, Record or Fixed Length** to open an on-screen menu of related Nuendo commands. The upper and lower row buttons fire the entries (the menu stays open so you can fire several); press any mode button (Mix, Clip, Device…) to leave. A short tap keeps each button's normal action.

Button	Menu entries
New	Track Version, Events to Part, Real Copy, Render in Place, Group, Insert Marker, Bounce
Duplicate	Duplicate Track, Duplicate Version, Duplicate Clip, Repeat, Fill Loop
Mute	Unmute All, Bypass Sends / Inserts / Channel Strip / EQs
Record	Audio/MIDI/MIDI-Cycle record modes, Retrospective Record, Punch In/Out, Punch=Cycle; upper row: Pre-roll, Post-roll
Play	Play Selection (solo), From Start / Until End, To Marker, Shuttle 2x / ½x; upper row: Start Mode, Auto RTZ, Until Start, From End, Preview, Set Left/Right Locator
Fixed Length	Loop Selection, Loop Selection Solo

> ***Note — Bypass commands:*** *Nuendo only applies the Mixer → Bypass: Sends / Inserts / Channel Strip / EQs key commands when the* ***MixConsole window is focused****. Click into the MixConsole first (or keep it focused on a second screen), then fire the bypass entries from the Push.*

7. Mixer Modes

Volume Mode (Mix button)

Shows 8 tracks with volume bars, VU meters, and peak clip indicators. Turn an encoder to adjust volume.

Upper row buttons (above the display):

- Short press = select track
- Shift + press = open Edit Channel Settings for that track
- Long press (1 second) = open instrument UI
- Double press = open Edit Channel Settings

Lower row buttons (below the display):

- Toggle Mute, Solo, Monitor, or Record Arm on the corresponding track
- Press the Mute button to switch between Mute and Monitor mode
- Press the Solo button to switch between Solo and Record Arm mode
- Long press (0.5s) in Mute mode = toggle Monitor on that track
- Long press (0.5s) in Solo mode = toggle Record Arm on that track
- Shift+Mute = clear all mutes (or all monitors in Monitor mode)
- Shift+Solo = clear all solos (or all rec arms in Rec mode)

  ***Nuendo 15 Group Folders**: Mute, Solo, Monitor and Record work on Group Folder tracks too. Monitor and Record act on the folder track through Nuendo key commands, so pressing them **selects the folder** in Nuendo. The Push LEDs show what you toggled from the Push, not changes made in Nuendo.*

Pan Mode (Shift+Clip)

Shows 8 tracks with pan position indicators. Turn an encoder to adjust pan.

Track Mode (Shift+Mix)

Combined mode for the selected track: Enc1=Volume, Enc2=Pan, Enc3-8=Sends 1-6.

8. Sends Mode

Press Clip. Encoders = send levels. Upper row = On/Off. Lower row = Pre/Post. Left/Right = navigate tracks.

9. Inserts Mode

List View: Short press upper = params. Long press = open UI. Shift+upper = params+UI. Lower = bypass. Shift+Left/Right = slots 1-8 / 9-16.

Parameter View: Encoders = 8 params. Left/Right = bank/page nav. Lower 1=UI, 2=bypass, 3=deactivate. If mapped: ★ MAPPED indicator.

10. Plugin Browser

> *Requires Nuendo 15+ (MIDI Remote API 1.3)*

Press Add Device to open the browser.

Phase 1 — Slot Selection

The screen shows the insert slots of the selected track.

- Upper row buttons (above the display) = select the target slot
- Left/Right = switch between slots 1-8 and 9-16
- Lower row button 8 = Cancel

Phase 2 — Folders

The collection is presented as a list of folders. **All** (top of the list) shows the whole collection flat.

- Encoder = scroll the folder list
- **OPEN** (lower row 1, or any upper-row button) = enter the highlighted folder
- **COLL ■** (lower row 2) = choose a different plugin collection
- **BACK** (lower row 8) = back to slot selection

  *Folders come from your Nuendo Plugin Manager collection structure (mCollectionPath). For the Default collection you can switch between grouping by **Type** or by **Vendor** — see Setup → Browser.*

Phase 3 — Plugin List

A vertical list showing each plugin's **name**, **type** and **manufacturer**.

- Encoder 1 = page, Encoder 2 = fine (moves the highlight)
- **LOAD** (lower row 1, or any upper-row button) = load the highlighted plugin
- **■ FOLDERS** (lower row 2) / **BACK** (lower row 8) = back to the folder list

After loading, the bridge returns to Inserts mode and rescans slots automatically.

> *The plugin list is **cached to disk**, so it opens instantly after the first load (a background refresh keeps it current). The bridge also remembers your **last-used collection** and reopens it next time.*

11. Quick Controls (Device Mode)

Press Device. 8 Quick Controls of the selected track. Left/Right = navigate tracks.

12. Transport Controls

Button	Function	LED
Play	Playback	White/Green/Purple
Record	Recording	Orange/Blink Red/Solid Red

Fixed Length	Cycle/Loop	
Automate	Automation cycle	White/Green/Red
Metronome	Metronome	LED reflects state
Shift+Metronome	Count-in (precount)	Blinks: slow = metronome off, fast = metronome on

- **Count-in LED**: Nuendo does not report the count-in state to MIDI Remote scripts, so the bridge only tracks its own Shift+Metronome presses. If you toggle the count-in in Nuendo itself, the bridge cannot see that change and the LED will be out of step — toggle it once more in Nuendo to realign.

13. Automation

Automate button cycles: Off → Read → Read+Write → Write → Off.

Holding values while writing (plateaus)

- **Encoders are touch-sensitive**: in **Volume**, **Pan** and **Quick Controls** modes, keep your finger resting on an encoder after turning it and the current value is held for the automation write — release the encoder to let the automation return. Several encoders can be held at once.
- **Send levels and plugin parameters** use relative encoders and cannot be held this way — to ride a plugin parameter with plateaus, assign it to a **Quick Control** and ride it from Device mode (or with the expression pedal's QC target).
- The **touchstrip** (Volume mode) holds its value the same way while your finger stays on the strip.
- The **expression pedal** (Volume / QC / Send targets) holds its last position while the transport is rolling — park the pedal during a pass and the value stays written until you stop.

14. Touchstrip

Shift+Layout to cycle: Pitch Bend, Mod Wheel (CC 1), Volume.

- In **Volume** mode the LED bar mirrors the selected track's actual volume — it follows fader moves made in Nuendo, track selection, and automation playback.

15. Note Input

64 pads: Chromatic (default) or Drum (Layout). Scale, Accent, Note Repeat available. Set instrument input to BridgeNotes.

Chord display

Hold two or more pads and the chord name appears on screen — in the footer of the Mix and Pan pages (in place of the pad note range), or as a green badge in the top-right corner of every other page. It stays visible for a moment after you release the pads.

- Recognised: major, minor, dim, aug, sus2, sus4, power chords (“C5”), 6, m6, 7, maj7, m7, mMaj7, m7b5, dim7, aug7, 7sus4, add9, m(add9), 6/9, 9, maj9, m9, 7b9, 7#9, 11, m11, 13, plus seventh / ninth voicings without the fifth.

- The lowest note is tried as the root first (C-E-G-A reads C6, A-C-E-G reads Am7); otherwise the chord is shown as an inversion (“C/E”).
- Sharps or flats follow the active scale (e.g. F major → Bb, E minor → F#). When the scale doesn't decide — Piano layout, Chromatic, Whole Tone, Octatonic, Spanish, or keys without accidentals such as C major — the preference in Setup → MIDI Ctrl is used.
- Only pads you hold are counted (not notes played back by Nuendo or held by the sustain pedal). Keyswitch pads and the Drum layout are ignored.

Scale selector (Scale button): pick a musical scale or the **Piano** layout. Piano mode arranges the pads like a piano keyboard — white keys on the lower row of each octave pair, black keys on the upper row, four octaves stacked bottom-to-top, every C in purple. The root is locked to C in Piano mode; use Octave Up/Down to move the whole four-octave block.

16. Control Room

Press User. 4 pages: Main, Phones, Cues, Sources. Master encoder = Main level. User+Master = Phones.

17. Channel Strip

> *Extended-parameter and EQ-curve control require Nuendo 15+ (API 1.3).*

From the Mixer view, press **Mix again** to enter Channel Strip mode. The overview shows the six strip sections as Cubase-coloured banners (Gate, Comp, EQ, Tools, Sat, Limiter) plus PreFilter Phase and PreGain.

> ***Important:*** *a strip module can only be controlled once it has been* ***activated manually in Nuendo****. Open the channel's strip in the Nuendo mixer and load/enable each module (Gate, Comp, EQ, Tools, Sat, Limiter) you want to use. An empty or disabled strip slot is not exposed to the Push 2 — its banner and encoders stay inactive until you turn the module on in Nuendo.*

Overview

- Each banner shows the section colour (dimmed when bypassed) and the loaded plugin variant.
- Lower row 1-6 = Bypass each section (pill/LED amber when bypassed).
- Lower row 7 = PreFilter Phase On/Off. Lower row 8 = Edit Channel Settings (white when the window is open).
- Upper row 1-6 = drill into that module's sub-page.

Module Sub-pages

- 8 encoders control that module's parameters. Parameters beyond Cubase's bank zone are exposed via DirectAccess (e.g. VintageCompressor Ratio & Mix, Tube Compressor Character, DeEsser side-chain filter, Maximizer Release/Recover).
- Lower row = the module's binary toggles (Auto Release, Solo, Diff, etc.) — LEDs follow the on-screen pills.
- Upper row layout: 1 = module name (inactive), 2 = Bypass, 3-5 = variant chips, 8 = back to overview.

  > *Variant chips show the currently loaded plugin. Switching the variant must be done in Nuendo's strip-slot menu — the MIDI Remote API does not allow changing strip-slot plugins from a controller; the chip mirrors the active variant in real time.*

18. Channel EQ Page

Drill into the EQ section for an EQ-Eight-style page with a live frequency-response curve.

Encoder	Function
1	Band selector (1-4) — turn to pick the active band
2	Filter Type of the selected band (bands 2-3 are fixed Peak)
3	Frequency of the selected band
4	Q of the selected band
5	Gain of the selected band
6	PreFilter Low-Cut frequency

7	PreFilter High-Cut frequency
8	PreGain

- Lower row 1 = selected band On/Off. Lower 6 = LC On, Lower 7 = HC On, Lower 8 = PreFilter bypass.
- Upper row 2 = EQ section bypass (amber when bypassed). Upper 8 = back.
- The curve combines the 4 EQ bands + PreFilter HC/LC. Numbered circles mark each band at its (freq, gain) point — white when the band is on, grey when off. The curve and markers update live, whether you change a parameter from the Push or directly in Nuendo.

19. Setup Page

Press Setup. The upper-row buttons switch tabs: **MIDI Ctrl**, **Vel Curve**, **CR Knob**, **Pedal 1**, **Pedal 2**, and **About**. Upper-row button 7 is **Reload Script** (re-scan tracks and re-sync the bridge's view of the project), available on every tab. All Setup choices are saved to `~/.push2bridge/settings.json` and restored on the next launch.

MIDI Ctrl

- Aftertouch mode for the pads (lower 1-3): Polyphonic (per-note pressure), Channel (global pressure), or Off.
- **Chord names** (lower 5 = *Chord #*, lower 6 = *Chord b*): sharps or flats for the chord display when the active scale doesn't decide (see Note Input).

Vel Curve

- Five velocity-response presets (Linear, Log, Exp, S-Curve, Fixed). In Fixed mode, Encoder 5 sets the fixed velocity value.
- **Encoder 6 — Sens** (0-10, default 5): how hard you must hit to reach full velocity. Raise it if the top of the range is hard to reach, lower it if full velocity comes too easily.
- **Encoders 7 / 8 — Min / Max**: the velocity sent by the softest and the hardest hit; the chosen curve is spread between them. The previews follow the settings live.

  Sens, Min and Max apply to Linear, Log, Exp and S-Curve; Fixed ignores them (they are greyed out).

CR Knob

- Sets what the Master encoder controls by default — Control Room *Main* or *Phones* level. Holding *Select* always gives the other one.

Pedal 1 / Pedal 2 (Footswitch)

- The Push 2 has two pedal jacks: jack 1 sends CC 64, jack 2 sends CC 69. Each pedal has its own tab. The lower row picks the action — *Sustain* (CC 64 passed through to the instrument), *Play/Stop*, *Play*, *Expr* (expression pedal, below), *Rec*, *Rec/Stop* (start recording, press again to stop), or *Off* — and lower-row button 8 toggles *Invert* (polarity: normally-closed switches, or heel/toe direction for an expression pedal). Defaults: Pedal 1 = Sustain, Pedal 2 = Play/Stop.

Expression pedal (Expr)

- Plug a passive expression pedal (TRS) into either jack and select **Expr**. Then **press Expr a second time to calibrate**: move the pedal over its full range while the screen shows the live readings, and press Expr again to finish. The bridge detects the pedal's wiring automatically (tip or ring) and stores the calibration — it is reapplied on every launch, so you only calibrate once per pedal.
- **Target** (Encoder 7) selects what the pedal controls; **Encoder 8** picks the parameter where applicable:
- **MIDI CC** — continuous CC to the instrument (Encoder 8 picks the CC, default 11 Expression).

- **Volume** — volume of the selected track (the touchstrip LED bar follows).
- **QC** — a Quick Control of the selected track (Encoder 8 picks QC 1-8).
- **Send** — a send level of the selected track (Encoder 8 picks Send 1-8).
- **AI Knob** — controls whatever parameter the mouse pointer is hovering in Nuendo, like Steinberg's AI Knob.

  *The AI Knob target is a relative control aimed at a moving target (whatever the mouse hovers), so it cannot hold plateau values for automation writing the way the other targets do. To ride a specific plugin parameter with plateaus, assign that parameter to a **Quick Control** in Nuendo and use the **QC** target instead.*

Browser

- Controls how the **Default** plugin collection is grouped into folders in the Plugin Browser: by **Type** (the Default collection's own type folders) or by **Vendor** (virtual folders per manufacturer). This setting applies to the Default collection only — every other collection always keeps its own folders. The *All* (flat) folder is available in both modes.

20. Navigation Page

Press the **Master** button to overlay four directional D-pads on the 64 pads, one in each corner: **Zoom** (top-left), **Scroll / cursor** (top-right), **Markers / locators** (bottom-left) and **Nudge** (bottom-right). Each pad has hold-to-repeat. The overlay stays active across the mix pages, so you can navigate the timeline while you keep working — and Note, Scale or Session restore the normal pads.

Showing the legend

- **Hold the Master button** to flash an on-screen legend of the four D-pads (what each direction does). Release Master and the screen returns to the mix while the D-pads stay live. Tap Master any time you need a reminder.

  The D-pads drive Nuendo Key Commands via the MIDI Remote script. If a direction does nothing, verify the matching command name in Nuendo's File → Key Commands (names must match your version / language).

21. MIDI CC Controller

Shift+Note. 8 assignable CC faders on BridgeNotes port. Upper row = edit CC number. Lower row = toggle 0/127. Defaults: CC 1,2,7,8,10,11,64,65.

22. Plugin Mapper

Create custom parameter mappings for your plugins. Access via the macOS menu bar / Windows tray (Plugin Mapper) or http://localhost:8100. Plugins the scanner can't read (iLok/PACE, Waves shells, Nuendo stock plug-ins) can be captured live from Nuendo with **Shift+Browse**; the **Edit Map** button (Inserts parameter page) opens the Mapper on the selected plugin. Mappings are saved in **~/.push2bridge/mappings/** — after saving, press ■ **Reload** (Setup) to pick them up. See the separate Plugin Mapper Guide for details.

All Mapper dependencies are bundled in the standalone .app/.exe — nothing to install. Only source installs need: pip install fastapi uvicorn pedalboard.

23. Key Commands

The eight note-value buttons (1/32t … 1/4, left of the pads) can trigger any Nuendo Key Command. There are **8 pages** of 8 slots — use **Page Left / Right** to switch pages; the current page number shows in the Mix footer.

- **Long-press** any note-value button to open the on-screen **legend** for the current page (slot number, printed button label, assigned command). It closes when you press a screen-changing button.
- Each button's colour is set per slot in the editor; **white is reserved for Note Repeat** when Repeat is engaged.

Configuring

- The editor lives inside the **Plugin Mapper** web page. Open **Plugin Mapper** from the macOS menu bar / Windows tray (or go to http://localhost:8100), then click the ■ **Key Commands** → link at the top of the page (direct URL: http://localhost:8100/keycmd). Pick a page, assign each slot a Nuendo function (searchable list), set a colour and an optional custom label.
- The function list is populated from your Nuendo Key Commands: in Nuendo, **File** → **Key Commands** → **Export**, then **Import Functions** in the editor. Names must match your Nuendo version / language.
- Press **Generate & install** — the bridge bakes the bindings into the installed Ableton_Push2.js and updates it in place.

  *After Generate & install, reload **both** sides: **reload the MIDI Remote script in Nuendo** so the new command bindings take effect, and on the Push press **Setup** → ■ **Reload** (or restart the bridge) so it picks up the new colours, labels and legend.*

  *macOS: the app needs permission to reach ~/Documents to find and update the installed script. If Generate & install reports it can't find the script, add the app to **System Settings** → **Privacy & Security** → **Full Disk Access**, then quit and relaunch.*

24. Button Reference

Button	Action	Shift + Button
Mix	Volume mode (press again: Channel Strip)	Track mode
Clip	Sends mode (press again: Pan)	
Device	Quick Controls	
Browse	Inserts mode	Capture params (DirectAccess)
Add Device	Plugin Browser	
Session	XY pad	
Master	Navigation D-pads	

Select	(hold) Master encoder → Phones	
Note	MIDI note pads	MIDI CC controller
Setup	Setup page	
Left/Right	Bank nav (8) / track nav	Nudge (1) in Vol/Pan
Play	Playback	
Record	Record toggle	
Mute	Mute/Monitor	Clear all
Solo	Solo/Rec arm	Clear all
User	Control Room	
Layout	Drum/Chromatic	Touchstrip cycle
Upper (dbl / Shift)	Edit Channel Settings	
Scale	Scale selector	
Accent	Fixed velocity	
Repeat	Auto-repeat	
Metronome	Metronome on/off	Count-in on/off

25. MIDI Channel Allocation

Channel	Usage
1 (0xB0)	Mixer (volume, pan, transport, selection, VU)
2 (0xB1)	Insert plugin parameters
3 (0xB2)	Send levels (selected track)
4 (0xB3)	Insert bypass toggles
5 (0xB4)	Quick Controls
6 (0xB5)	Control Room
7 (0xB6)	Bank zone sends
8 (0xB7)	DirectAccess commands (browser, bypass, edit, strip slot, param-flip, variant)
9 (0xB8)	DirectAccess encoder control (mapped params, strip slots)
16 (0xBF)	Heartbeat

26. Troubleshooting

- **Push 2 not found**: check USB, libusb, close Ableton Live.
- **push2-python error**: install it from source — pip install git+https://github.com/ffont/push2-python.git. Use Python 3.11.9 or lower.
- **No Nuendo connection**: check bridge is running; add the surface manually (MIDI Remote Manager → Add, Vendor Ableton / Model Push2).
- **Pads play doubled notes**: uncheck “In ‘All MIDI Inputs’” for the Push 2 Live Port and User Port (Studio Setup → MIDI Port Setup).
- **Channel Strip / EQ curve not working**: requires Nuendo 15+ (API 1.3). Drill into a section once to trigger DA strip exploration.
- **Plugin Mapper not loading**: pip install fastapi uvicorn pedalboard.
- **Track names not loading**: navigate Left/Right to refresh.
- **MIDI tracks change parameters on their own**: uncheck “In ‘All MIDI Inputs’” for **NuendoBridge Out** (Studio Setup → MIDI Port Setup).
- **Metronome button blinks but the count-in is off (or the reverse)**: the count-in was changed in Nuendo, which the bridge cannot see — toggle it once more in Nuendo to realign.
- **Hidden/shown tracks not reflected on the Push**: after changing track visibility in Nuendo, the display updates by itself within ~10-20 seconds — **select any track** in Nuendo to refresh it immediately.

  Logs: macOS — ~/Library/Logs/Push2NuendoBridge.log. Windows — terminal output.

27. Backup & Restore

All of your bridge settings live in **~/.push2bridge/** (Setup options, footswitch/pedal calibration, Key Commands, plugin mappings, last-used collection, Plugin Mapper settings). You can back them up and restore them as a single zip.

- From the macOS menu bar / Windows tray: **Backup Settings…** saves a zip; **Restore Settings…** loads one back.
- The Plugin Mapper web page also has ■ **Backup all** / ■ **Restore** buttons.

  Restoring overwrites the current settings with the ones in the zip. Restart the bridge (and Reload the script) after a restore.

28. Support

Push 2 / Nuendo Bridge is donationware.

- GitHub: https://github.com/mbourque-mix/Push2Nuendo-Bridge
- Buy Me A Coffee: https://buymeacoffee.com/mbourque
- Issues: https://github.com/mbourque-mix/Push2Nuendo-Bridge/issues

  Licensed under GPL-3.0. Built with push2-python and the Steinberg MIDI Remote API.